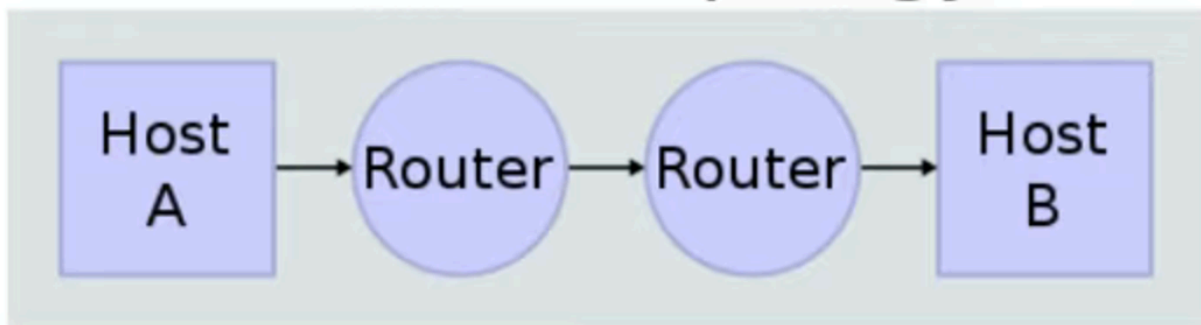


Day 30: TCP & UDP

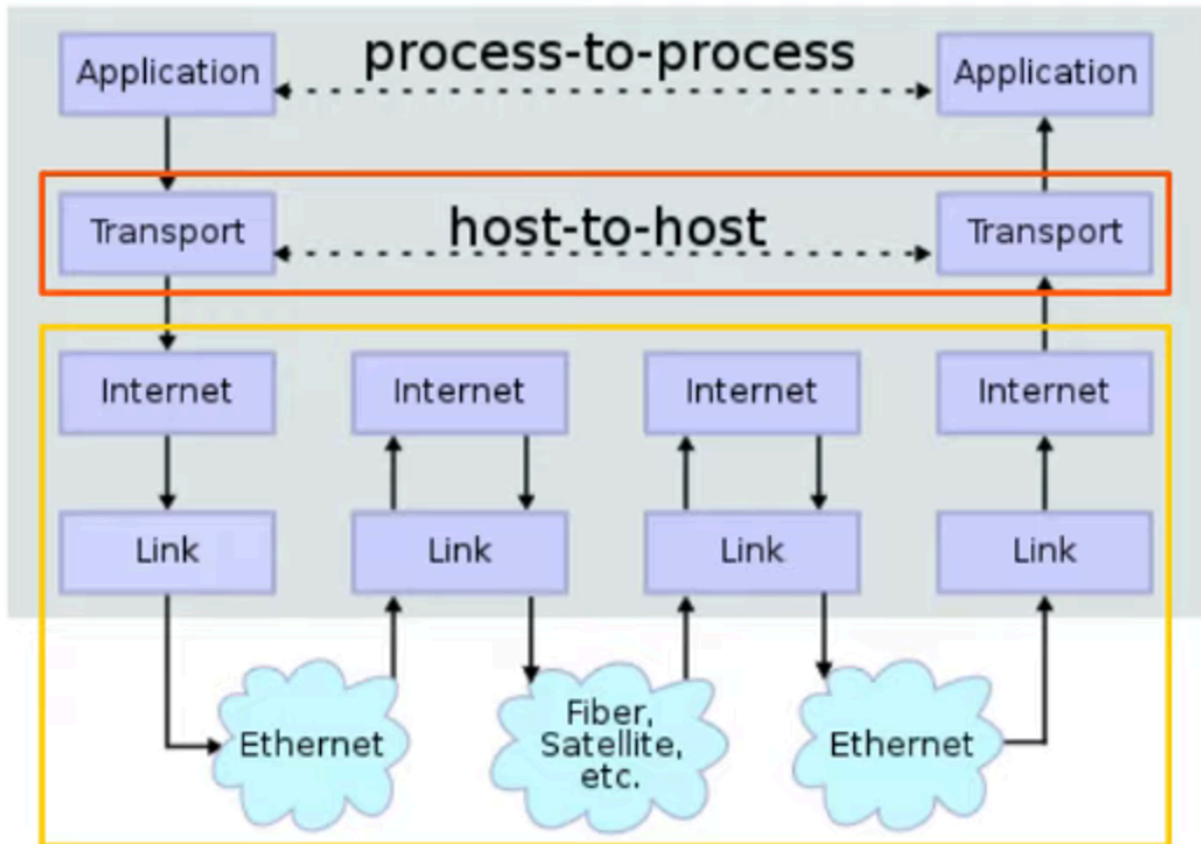
BASICS OF LAYER 4

- Provides TRANSPARENT transfer of DATA between END HOSTS (Host To Host communication)

Network Topology



Data Flow

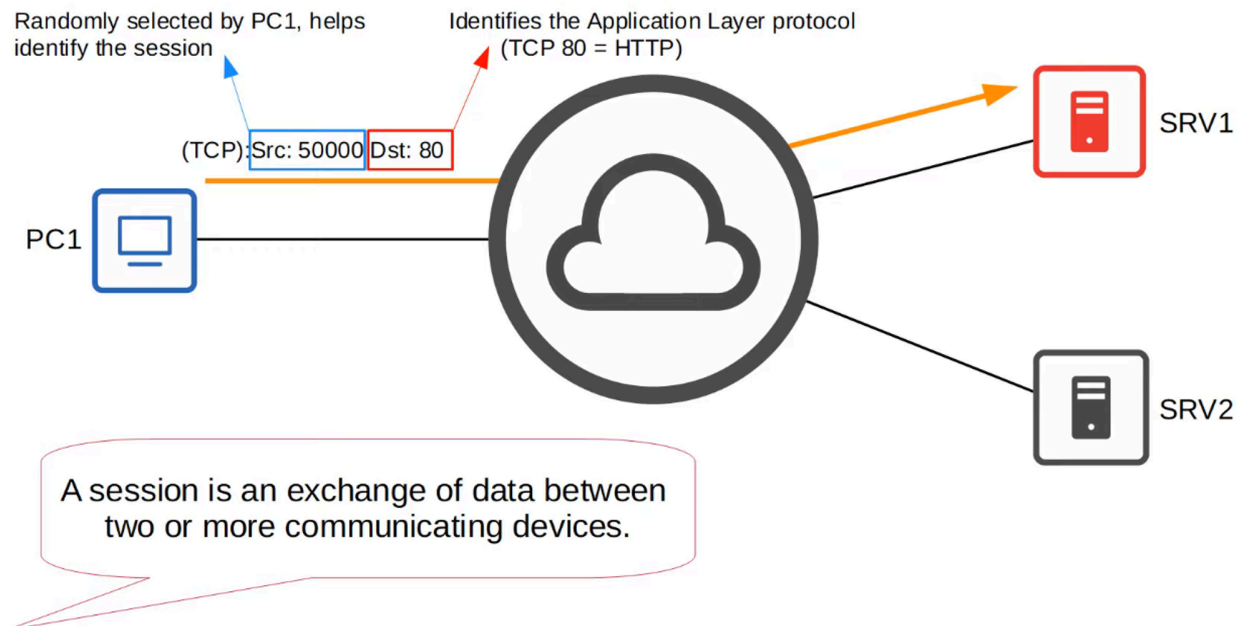


- Provides (or DOESN'T provide) various SERVICES to APPLICATIONS:
 - Reliable DATA Transfer
 - Error Recovery

- Data Sequencing
- Flow Control
- Provides LAYER 4 ADDRESSING (PORT numbers) - NOT the physical interfaces / ports on network devices
 - IDENTIFY the APPLICATION LAYER protocol
 - Provides SESSION multiplexing

WHAT IS A SESSION ?

- A SESSION is an EXCHANGE of DATA between TWO or MORE communicating DEVICES



The FOLLOWING ranges have been designated by IANA (Internet Assigned Numbers Authority)

- **Well-Known** Port Numbers : 0 - 1023
- **Registered** Port Numbers : 1024 - 49151
- **Ephemeral** / Private / Dynamic port numbers : 49152 - 65535

TCP Header

16 bits = 65536(2^{16}) available port numbers

		TCP segment header																															
Offsets	Octet	0								1								2								3							
Octet	Bit	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0	0	Source port																Destination port															
4	32	Sequence number																															
8	64	Acknowledgment number (if ACK set)																															
12	96	Data offset		Reserved 000		NS	CWR	ECE	URG	ACK	PSH	RST	SYN	FIN	Window Size																		
16	128	Checksum																Urgent pointer (if URG set)															
20	160	Options (if data offset > 5. Padded at the end with "0" bytes if necessary.)																															
...	...	...																															



TCP (TRANSMISSION CONTROL PROTOCOL)

- A CONNECTION-ORIENTED protocol
 - Before actually SENDING DATA to the DESTINATION HOST, the TWO HOSTS communicate to establish a CONNECTION. Once the CONNECTION is established, DATA exchange begins.

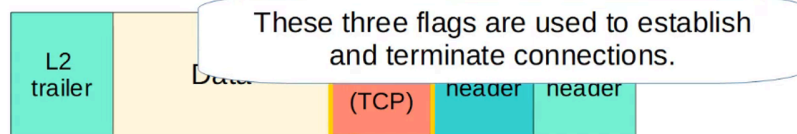
		TCP segment header																															
Offsets	Octet	0								1								2								3							
Octet	Bit	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0	0	Source port																Destination port															
4	32	Sequence number																															
8	64	Acknowledgment number (if ACK set)																															
12	96	Data offset	Reserved			NS		CWR	ECN	URG	ACK	PSH	RST	SYN	FIN	Window Size																	
16	128	Checksum																Urgent pointer (if URG set)															
20	160	Options (if data offset > 5. Padded at the end with "0" bytes if necessary.)																															
...	...																																

These two fields provide sequencing and reliable communication

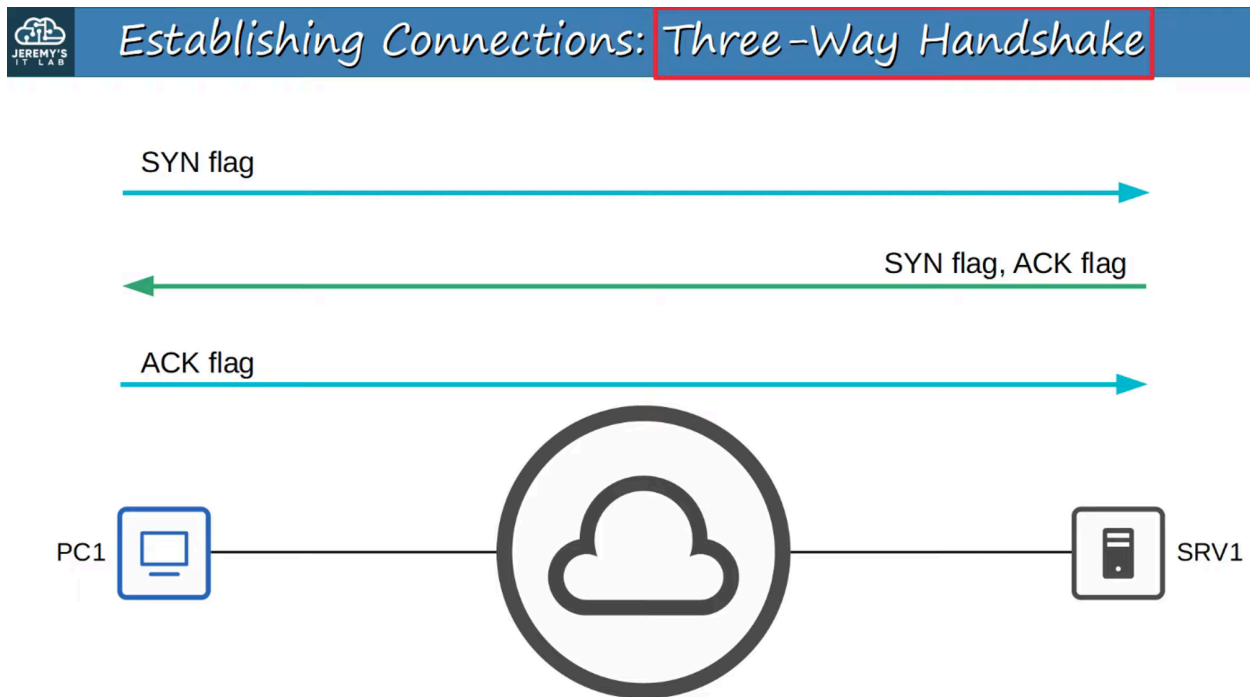
These two fields provide sequencing and reliable communication.



TCP segment header																																	
Offsets	Octet	0								1								2								3							
Octet	Bit	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0	0	Source port																Destination port															
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20	160	Options (if data offset > 5. Pad to the end with "0" bytes if necessary.)																															
...	...																																

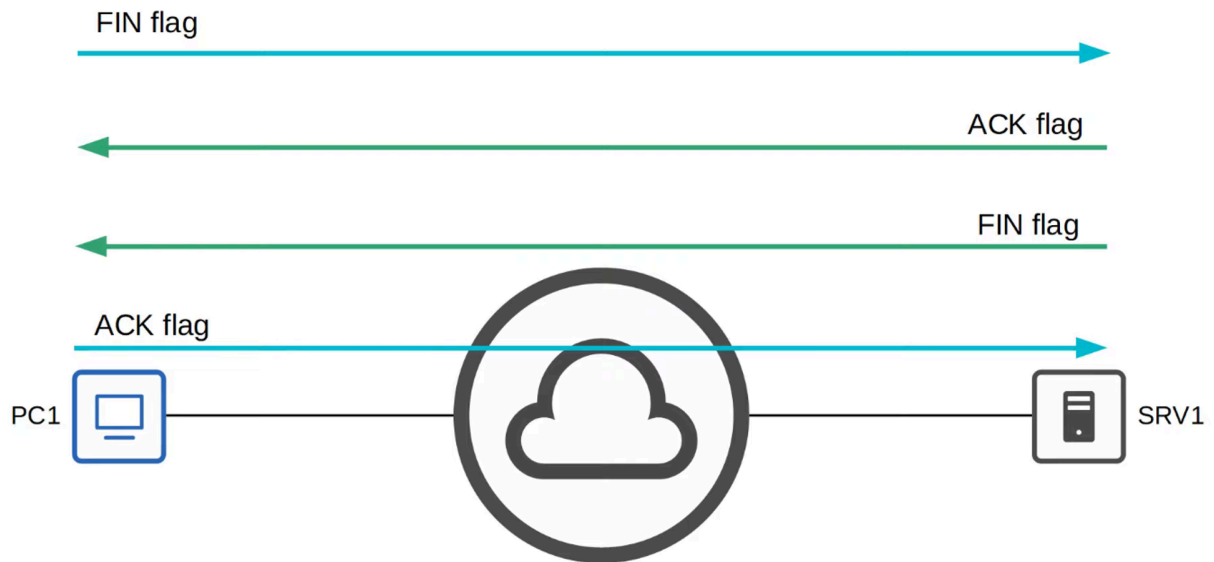


Establishing connections



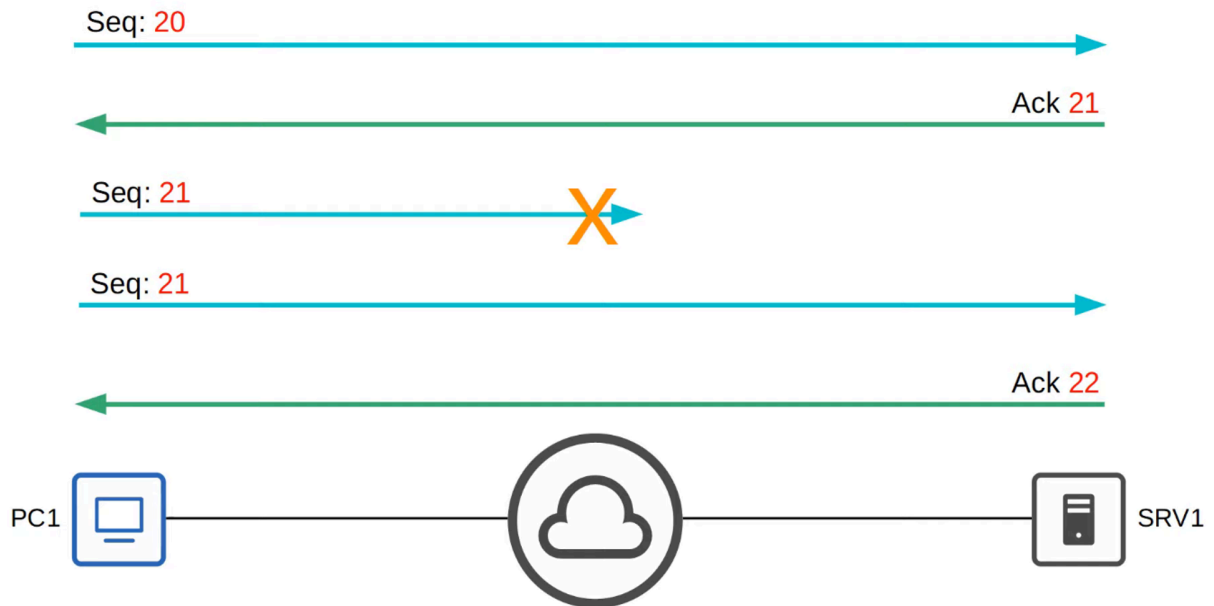
Terminating connections

Terminating Connections: Four-Way Handshake



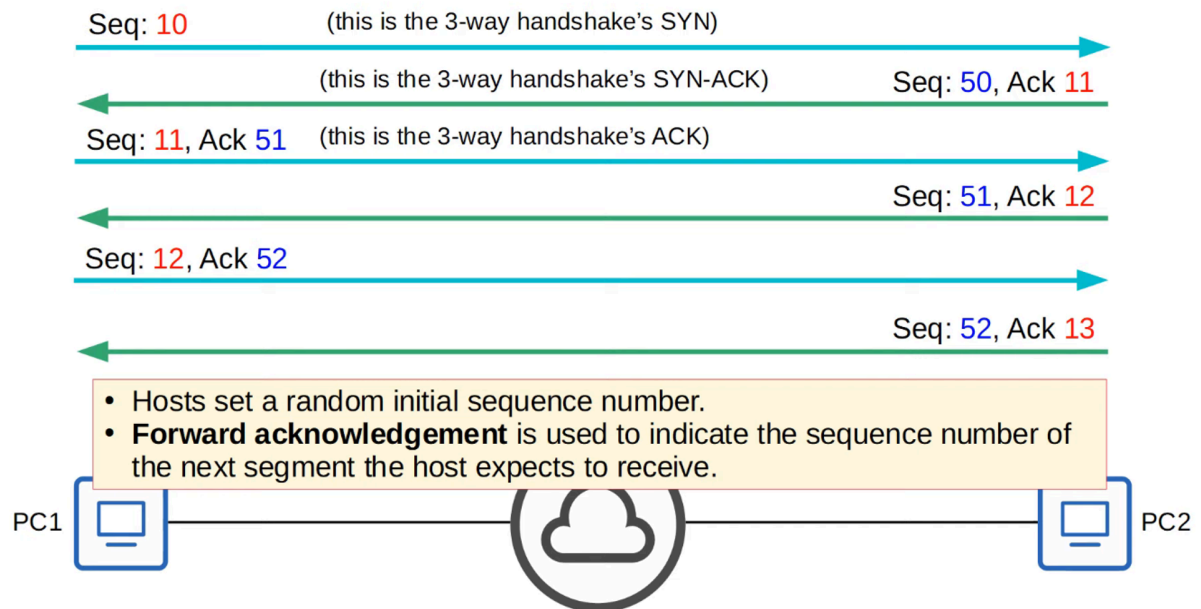
- TCP provides RELIABLE communication
 - The DESTINATION HOST must acknowledge that it RECEIVED each TCP SEGMENT (Layer 4 PDU)
 - If a SEGMENT isn't ACKNOWLEDGED, it is sent again

TCP Retransmission

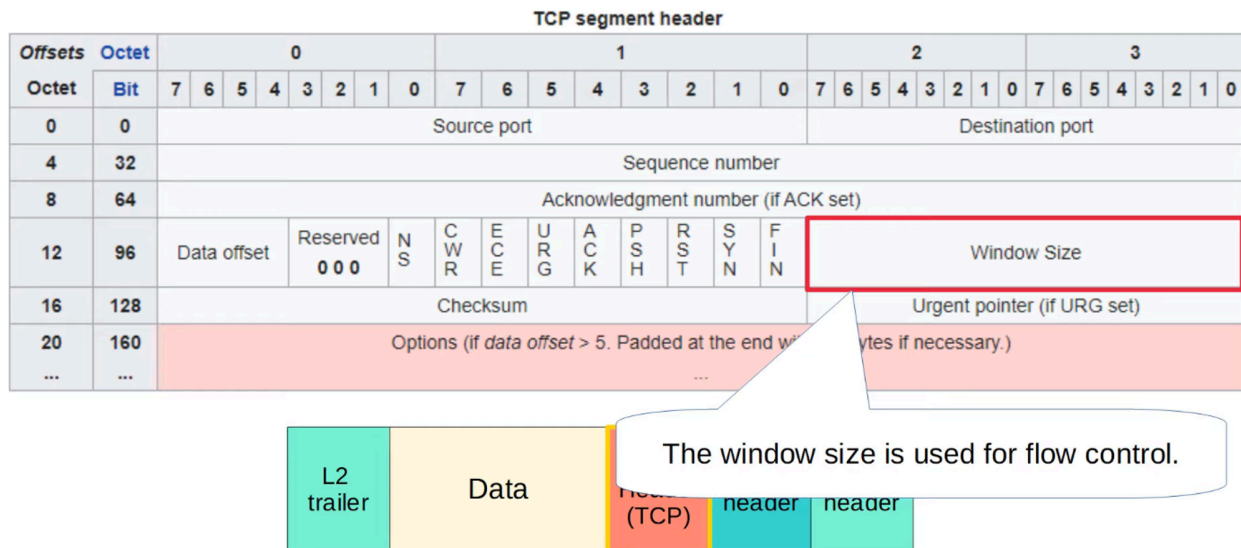


- TCP provides SEQUENCING
 - SEQUENCE numbers in the TCP HEADER allow DESTINATION HOSTS to put SEGMENTS in the correct ORDER even if they arrive out of ORDER

TCP: Sequencing / Acknowledgment



- TCP provides FLOW CONTROL
 - The DESTINATION HOST can tell the SOURCE HOST to increase / decrease the RATE that DATA is sent



TCP Flow Control: Window Size

- Acknowledging every single segment, no matter what size, is inefficient.
- The TCP header's **Window Size** field allows more data to be sent before an acknowledgment is required.
- A 'sliding window' can be used to dynamically adjust how large the window size is.

Seq: 20 →

Seq: 21 →

Seq: 22 →

← Ack 23

In all of these examples, I used very simple sequence numbers. In real situations, the sequence numbers get much larger and do not increase by 1 with each message. For the CCNA, just understand the concepts and don't worry about the exact numbers.

UDP (USER DATAGRAM PROTOCOL)

UDP datagram header																																	
Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Source port																Destination port															
4	32	Length																Checksum															

- UDP is NOT a CONNECTION-ORIENTED PROTOCOL
 - The SENDING HOST does NOT establish a CONNECTION with the DESTINATION HOST before sending DATA. The DATA is simply SENT
- UDP DOES NOT provide reliable COMMUNICATION
 - When UDP is used, ACKNOWLEDGEMENTS are NOT SENT for received SEGMENTS
 - If a SEGMENT is LOST, UDP has no mechanism to re-TRANSMIT it
 - SEGMENTS are sent "best-effort"
- UDP DOES NOT provide SEQUENCING
 - There is NO SEQUENCE NUMBER FIELD in the UDP header
 - If SEGMENTS arrive out of order, UDP has no MECHANISM to put them back in ORDER
- UDP DOES NOT provide FLOW CONTROL
 - UDP has NO MECHANISM like TCP's WINDOW SIZE to control the flow of DATA
- UDP DOES provide ERROR CHECKING (via CHECKSUM)

COMPARING TCP AND UDP

Number of Fields in their Headers

Comparing TCP & UDP

TCP segment header																																	
Offsets	Octet	0								1								2								3							
Octet	Bit	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0	0	Source port																Destination port															
4	32	Sequence number																															
8	64	Acknowledgment number (if ACK set)																															
12	96	Data offset				Reserved 000		N S	C W R	E C E	U R G	A C K	P S H	R S T	S Y N	F I N	Window Size																
16	128	Checksum																Urgent pointer (if URG set)															
20	160	Options (if data offset > 5. Padded at the end with "0" bytes if necessary.)																															
...	...	...																															

UDP datagram header																																	
Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Source port																Destination port															
4	32	Length																Checksum															

- TCP provides MORE FEATURES than UDP but at a COST of ADDITIONAL OVERHEAD
- For applications that require RELIABLE communications (for example, downloading a file), TCP is PREFERRED
- For applications, like real-time voice and video, UDP is preferred
- There are SOME applications that use UDP, but provide RELIABILITY, etc. within the APPLICATION itself.
- Some applications use BOTH TCP and UDP, depending on the situation.

TCP	UDP
Connection-oriented	Connectionless
Reliable	Unreliable
Sequencing	No sequencing
Flow control	No flow control
Use for downloads, file sharing, etc	Used for VoIP, live video, etc

IMPORTANT PORT NUMBERS

TCP

- FTP data (20)
- FTP control (21)
- SSH (22)
- Telnet (23)
- SMTP (25)
- HTTP (80)
- POP3 (110)
- HTTPS (443)

UDP

- DHCP server (67)
- DHCP client (68)
- TFTP (69)
- SNMP agent (161)
- SNMP manager (162)
- Syslog (514)

TCP & UDP

- DNS (53)